



Syllabus: Random variables, Uniform, Normal, Exponential, Poisson and Binomial distributions. Mean, median, mode and standard deviation. Conditional probability and Bayes theorem

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	1	0	2	1	0	0	1	1	0	0.75	2
2 Marks Count	1	2	1	1	1	0	2	2	0	1.25	2
Total Marks	3	4	4	3	2	0	5	5	0	3.25	5

7.1

Bayes Theorem (1)

7.1.1 Bayes Theorem: GATE DA 2025 | Question: 21



There are three boxes containing white balls and black balls.

Box -1 contains 2 black and 1 white balls.

Box- 2 contains 1 black and 2 white balls.

Box -3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box- 3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____ (Round off to two decimal places)

gateda-2025 probability bayes-theorem numerical-answers one-mark

Answer key

7.2

Bayesian Network (2)

7.2.1 Bayesian Network: GATE DS&AI 2024 | Question: 14



Consider five random variables U, V, W, X , and Y whose joint distribution satisfies:

$$P(U, V, W, X, Y) = P(U)P(V)P(W | U, V)P(X | W)P(Y | W)$$

Which ONE of the following statements is FALSE?

- A. Y is conditionally independent of V given W
- B. X is conditionally independent of U given W
- C. U and V are conditionally independent given W
- D. Y and X are conditionally independent given W

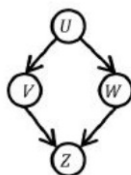
gate-ds-ai-2024 probability random-variable bayesian-network one-mark

Answer key

7.2.2 Bayesian Network: GATE DS&AI 2024 | Question: 54



Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	$P(\cdot)$
$U = 0$	0.5
$U = 1$	0.5

	$P(V = 0 \mid \cdot)$	$P(V = 1 \mid \cdot)$
$U = 0$	0.5	0.5
$U = 1$	0.5	0.5

	$P(W = 0 \mid \cdot)$	$P(W = 1 \mid \cdot)$
$U = 0$	1	0
$U = 1$	0	1

		$P(Z = 0 \mid \cdot)$	$P(Z = 1 \mid \cdot)$
$V = 0$	$W = 0$	0.5	0.5
$V = 0$	$W = 1$	1	0
$V = 1$	$W = 0$	1	0
$V = 1$	$W = 1$	0.5	0.5

The value of $P(U = 1, V = 1, W = 1, Z = 1) = \underline{\hspace{2cm}}$ (rounded off to three decimal places).

gate-ds-ai-2024 probability bayesian-network numerical-answers two-marks

Answer key 

7.3

Bernoulli Distribution (1)

7.3.1 Bernoulli Distribution: GATE DA 2025 | Question: 30



A random variable X is said to be distributed as $\text{Bernoulli}(\theta)$, denoted by $X \sim \text{Bernoulli}(\theta)$, if

$$P(X = 1) = \theta, \quad P(X = 0) = 1 - \theta$$

for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim \text{Bernoulli}(\theta)$, $i = 1, 2, \dots, 300$ be independent and identically distributed random variables with $\theta = 0.25$. The value of $P(60 \leq Y \leq 90)$, after approximation through Central Limit Theorem, is given by (Recall that $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$)

- A. $\phi(2) - \phi(-2)$
C. $\phi(3) - \phi(-3)$

- B. $\phi(1) - \phi(-1)$
D. $\phi(90) - \phi(60)$

gateda-2025 probability random-variable bernoulli-distribution two-marks

Answer key 

7.4

Binomial Distribution (6)

7.4.1 Binomial Distribution: GATE CSE 2002 | Question: 2.16



Four fair coins are tossed simultaneously. The probability that at least one head and one tail turn up is

A. $\frac{1}{16}$

B. $\frac{1}{8}$

C. $\frac{7}{8}$

D. $\frac{15}{16}$

Answer key

7.4.2 Binomial Distribution: GATE CSE 2005 | Question: 52



A random bit string of length n is constructed by tossing a fair coin n times and setting a bit to 0 or 1 depending on outcomes head and tail, respectively. The probability that two such randomly generated strings are not identical is:

- A. $\frac{1}{2^n}$ B. $1 - \frac{1}{n}$ C. $\frac{1}{n!}$ D. $1 - \frac{1}{2^n}$

gatecse-2005 probability binomial-distribution easy

Answer key

7.4.3 Binomial Distribution: GATE CSE 2006 | Question: 21



For each element in a set of size $2n$, an unbiased coin is tossed. The $2n$ coin tosses are independent. An element is chosen if the corresponding coin toss was a head. The probability that exactly n elements are chosen is

- A. $\frac{{}^{2n}C_n}{4^n}$ B. $\frac{{}^{2n}C_n}{2^n}$ C. $\frac{1}{{}^{2n}C_n}$ D. $\frac{1}{2}$

gatecse-2006 probability binomial-distribution normal

Answer key

7.4.4 Binomial Distribution: GATE IT 2005 | Question: 32



An unbiased coin is tossed repeatedly until the outcome of two successive tosses is the same. Assuming that the trials are independent, the expected number of tosses is

- A. 3 B. 4 C. 5 D. 6

gateit-2005 probability binomial-distribution expectation normal

Answer key

7.4.5 Binomial Distribution: GATE IT 2006 | Question: 22



When a coin is tossed, the probability of getting a Head is $p, 0 < p < 1$. Let N be the random variable denoting the number of tosses till the first Head appears, including the toss where the Head appears. Assuming that successive tosses are independent, the expected value of N is

- A. $\frac{1}{p}$ B. $\frac{1}{(1-p)}$ C. $\frac{1}{p^2}$ D. $\frac{1}{(1-p^2)}$

gateit-2006 probability binomial-distribution expectation normal

Answer key

7.4.6 Binomial Distribution: GATE IT 2007 | Question: 1



Suppose there are two coins. The first coin gives heads with probability $\frac{5}{8}$ when tossed, while the second coin gives heads with probability $\frac{1}{4}$. One of the two coins is picked up at random with equal probability and tossed. What is the probability of obtaining heads ?

- A. $\left(\frac{7}{8}\right)$ B. $\left(\frac{1}{2}\right)$ C. $\left(\frac{7}{16}\right)$ D. $\left(\frac{5}{32}\right)$

gateit-2007 probability normal binomial-distribution

Answer key

7.5 Conditional Probability (14)

7.5.1 Conditional Probability: GATE CSE 1994 | Question: 1.4, ISRO2017-2



Let A and B be any two arbitrary events, then, which one of the following is TRUE?

- A. $P(A \cap B) = P(A)P(B)$ B. $P(A \cup B) = P(A) + P(B)$
C. $P(A | B) = P(A \cap B)P(B)$ D. $P(A \cup B) \leq P(A) + P(B)$

gate1994 probability conditional-probability normal isro2017

Answer key

7.5.2 Conditional Probability: GATE CSE 1994 | Question: 2.6



The probability of an event B is P_1 . The probability that events A and B occur together is P_2 while the probability that A and B occur together is P_3 . The probability of the event A in terms of P_1, P_2 and P_3 is _____

gate1994 probability normal conditional-probability fill-in-the-blanks

Answer key

7.5.3 Conditional Probability: GATE CSE 2003 | Question: 3



Let $P(E)$ denote the probability of the event E . Given $P(A) = 1$, $P(B) = \frac{1}{2}$, the values of $P(A | B)$ and $P(B | A)$ respectively are

- A. $\left(\frac{1}{4}\right), \left(\frac{1}{2}\right)$ B. $\left(\frac{1}{2}\right), \left(\frac{1}{4}\right)$ C. $\left(\frac{1}{2}\right), 1$ D. $1, \left(\frac{1}{2}\right)$

gatecse-2003 probability easy conditional-probability

Answer key

7.5.4 Conditional Probability: GATE CSE 2005 | Question: 51



Box P has 2 red balls and 3 blue balls and box Q has 3 red balls and 1 blue ball. A ball is selected as follows: (i) select a box (ii) choose a ball from the selected box such that each ball in the box is equally likely to be chosen.

The probabilities of selecting boxes P and Q are $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Given that a ball selected in the above process is a red ball, the probability that it came from the box P is:

- A. $\frac{4}{19}$ B. $\frac{5}{19}$ C. $\frac{2}{9}$ D. $\frac{19}{30}$

gatecse-2005 probability conditional-probability normal

Answer key

7.5.5 Conditional Probability: GATE CSE 2008 | Question: 27



Aishwarya studies either computer science or mathematics everyday. If she studies computer science on a day, then the probability that she studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

- A. 0.24 B. 0.36 C. 0.4 D. 0.6

gatecse-2008 probability normal conditional-probability

Answer key

7.5.6 Conditional Probability: GATE CSE 2009 | Question: 21



An unbalanced dice (with 6 faces, numbered from 1 to 6) is thrown. The probability that the face value is odd is 90% of the probability that the face value is even. The probability of getting any even numbered face is the same. If the probability that the face is even given that it is greater than 3 is 0.75, which one of the following options is closest to the probability that the face value exceeds 3?

- A. 0.453 B. 0.468 C. 0.485 D. 0.492

gatecse-2009 probability normal conditional-probability

Answer key

7.5.7 Conditional Probability: GATE CSE 2011 | Question: 3



If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?

- A. $\left(\frac{1}{3}\right)$ B. $\left(\frac{1}{4}\right)$ C. $\left(\frac{1}{2}\right)$ D. $\left(\frac{2}{3}\right)$

gatecse-2011 probability easy conditional-probability

Answer key

7.5.8 Conditional Probability: GATE CSE 2012 | Question: 33



Suppose a fair six-sided die is rolled once. If the value on the die is 1, 2, or 3, the die is rolled a second time. What is the probability that the sum total of values that turn up is at least 6?

- A. $\frac{10}{21}$ B. $\frac{5}{12}$ C. $\frac{2}{3}$ D. $\frac{1}{6}$

gatecse-2012 probability conditional-probability normal

Answer key

7.5.9 Conditional Probability: GATE CSE 2016 Set 2 | Question: 05



Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____.

gatecse-2016-set2 probability conditional-probability normal numerical-answers

Answer key

7.5.10 Conditional Probability: GATE CSE 2017 Set 2 | Question: 26



P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$, the probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for this job is

- A. $\left(\frac{4}{5}\right)$ B. $\left(\frac{5}{6}\right)$ C. $\left(\frac{7}{8}\right)$ D. $\left(\frac{11}{12}\right)$

gatecse-2017-set2 probability conditional-probability

Answer key

7.5.11 Conditional Probability: GATE CSE 2018 | Question: 44



Consider Guwahati, (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and $P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D | H_G) = 0.40$. Similarly, the next two entries are $P(M_D | H_G) = 0.48$ and $P(L_D | H_G) = 0.12$. Similarly for the other rows.

If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$, and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

gatecse-2018 probability conditional-probability numerical-answers two-marks

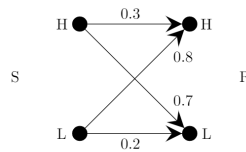
Answer key

7.5.12 Conditional Probability: GATE CSE 2021 Set 1 | Question: 54



A sender (S) transmits a signal, which can be one of the two kinds: H and L with probabilities 0.1 and 0.9 respectively, to a receiver (R).

In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the received signal is L given the transmitted signal was H , is 0.7.



If the received signal is H , the probability that the transmitted signal was H (rounded to 2 decimal places) is _____.

gatecse-2021-set1 probability conditional-probability numerical-answers two-marks

Answer key

7.5.13 Conditional Probability: GATE CSE 2024 | Set 1 | Question: 53



A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____.

gatecse2024-set1 numerical-answers probability conditional-probability two-marks

Answer key

7.5.14 Conditional Probability: GATE IT 2006 | Question: 1



In a certain town, the probability that it will rain in the afternoon is known to be 0.6. Moreover, meteorological data indicates that if the temperature at noon is less than or equal to 25°C , the probability that it will rain in the afternoon is 0.4. The temperature at noon is equally likely to be above 25°C , or at/below 25°C . What is the probability that it will rain in the afternoon on a day when the temperature at noon is above 25°C ?

- A. 0.4 B. 0.6 C. 0.8 D. 0.9

gateit-2006 probability normal conditional-probability

Answer key

7.6 Continuous Distribution (1)

7.6.1 Continuous Distribution: GATE CSE 2016 Set 1 | Question: 04



A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is _____.

gatecse-2016-set1 probability normal numerical-answers continuous-distribution

Answer key

7.7 Expectation (13)

7.7.1 Expectation: GATE CSE 1999 | Question: 1.1



Suppose that the expectation of a random variable X is 5. Which of the following statements is true?

- A. There is a sample point at which X has the value 5.
 B. There is a sample point at which X has value greater than 5.
 C. There is a sample point at which X has a value greater than equal to 5.
 D. None of the above.

gate1999 probability expectation easy

Answer key

7.7.2 Expectation: GATE CSE 2004 | Question: 74



An examination paper has 150 multiple choice questions of one mark each, with each question having four choices. Each incorrect answer fetches -0.25 marks. Suppose 1000 students choose all their answers randomly with uniform probability. The sum total of the expected marks obtained by all these students is

A. 0

B. 2550

C. 7525

D. 9375

gatecse-2004 probability expectation normal

Answer key

7.7.3 Expectation: GATE CSE 2006 | Question: 18



We are given a set $X = \{X_1, \dots, X_n\}$ where $X_i = 2^i$. A sample $S \subseteq X$ is drawn by selecting each X_i independently with probability $P_i = \frac{1}{2}$. The expected value of the smallest number in sample S is:

A. $(\frac{1}{n})$

B. 2

C. $\sqrt{n}$ D. n

gatecse-2006 probability expectation normal

Answer key

7.7.4 Expectation: GATE CSE 2011 | Question: 18



If the difference between the expectation of the square of a random variable ($E[X^2]$) and the square of the expectation of the random variable ($E[X]^2$) is denoted by R , then

A. $R = 0$ B. $R < 0$ C. $R \geq 0$ D. $R > 0$

gatecse-2011 probability random-variable expectation normal

Answer key

7.7.5 Expectation: GATE CSE 2013 | Question: 24



Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is $\frac{1}{2}$. What is the expected number of unordered cycles of length three?

A. $\frac{1}{8}$

B. 1

C. 7

D. 8

gatecse-2013 probability expectation normal

Answer key

7.7.6 Expectation: GATE CSE 2014 Set 2 | Question: 2



Each of the nine words in the sentence "The quick brown fox jumps over the lazy dog" is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The expected length of the word drawn is _____. (The answer should be rounded to one decimal place.)

gatecse-2014-set2 probability expectation numerical-answers easy

Answer key

7.7.7 Expectation: GATE CSE 2017 Set 2 | Question: 31



For any discrete random variable X , with probability mass function $P(X = j) = p_j, p_j \geq 0, j \in \{0, \dots, N\}$, and $\sum_{j=0}^N p_j = 1$, define the polynomial function $g_x(z) = \sum_{j=0}^N p_j z^j$. For a certain discrete random variable Y , there exists a scalar $\beta \in [0, 1]$ such that $g_y(z) = (1 - \beta + \beta z)^N$. The expectation of Y is

A. $N\beta(1 - \beta)$ B. $N\beta$ C. $N(1 - \beta)$ D. Not expressible in terms of N and β alone

gatecse-2017-set2 probability random-variable difficult expectation

Answer key

7.7.8 Expectation: GATE CSE 2021 Set 1 | Question: 35



Consider the two statements.

- S_1 : There exist random variables X and Y such that $(\mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))])^2 > \text{Var}[X]\text{Var}[Y]$
- S_2 : For all random variables X and Y , $\text{Cov}[X, Y] = \mathbb{E}[|X - \mathbb{E}[X]| |Y - \mathbb{E}[Y]|]$

Which one of the following choices is correct?

A. Both S_1 and S_2 are trueB. S_1 is true, but S_2 is false

C. S_1 is false, but S_2 is true

D. Both S_1 and S_2 are false

gatecse-2021-set1 probability random-variable difficult two-marks expectation

Answer key

7.7.9 Expectation: GATE CSE 2021 Set 2 | Question: 29



In an examination, a student can choose the order in which two questions (QuesA and QuesB) must be attempted.

- If the first question is answered wrong, the student gets zero marks.
- If the first question is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question.
- If both the questions are answered correctly, the student gets the sum of the marks of the two questions.

The following table shows the probability of correctly answering a question and the marks of the question respectively.

question	probability of answering correctly	marks
QuesA	0.8	10
QuesB	0.5	20

Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?

- A. First QuesA and then QuesB. Expected marks 14.
B. First QuesB and then QuesA. Expected marks 14.
C. First QuesB and then QuesA. Expected marks 22.
D. First QuesA and then QuesB. Expected marks 16.

gatecse-2021-set2 probability expectation two-marks

Answer key

7.7.10 Expectation: GATE DA 2025 | Question: 1



Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X | Y]$. Then $E[E[X | Y]]$ equals

- A. $E[X | Y]$ B. $\frac{E[X]}{E[Y]}$ C. $E[X]$ D. $E[Y]$

gateda-2025 probability random-variable expectation one-mark

Answer key

7.7.11 Expectation: GATE DA 2025 | Question: 10



Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4$$

where $E[X]$ denotes the expectation of random variable X . The values of a, b are:

- A. $a = -2, b = 1$ B. $a = 2, b = -1$ C. $a = -2, b = -1$ D. $a = 1, b = 1$

gateda-2025 probability expectation random-variable one-mark

Answer key

7.7.12 Expectation: GATE DA 2025 | Question: 51



A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment.

The expectation of S_{100} denoted by $E[S_{100}] = \underline{\hspace{2cm}}$. (Round off to one decimal place)

gateda-2025 probability expectation numerical-answers two-marks

Answer key

7.7.13 Expectation: GATE DS&AI 2024 | Question: 26



A fair six-sided die (with faces numbered 1, 2, 3, 4, 5, 6) is repeatedly thrown independently.

What is the expected number of times the die is thrown until two consecutive throws of even numbers are seen?

- A. 2 B. 4 C. 6 D. 8

gate-ds-ai-2024 probability expectation two-marks

Answer key

7.8 Exponential Distribution (5)

7.8.1 Exponential Distribution: GATE CSE 2021 Set 1 | Question: 18



The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2. For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is _____.

gatecse-2021-set1 probability random-variable numerical-answers one-mark exponential-distribution

Answer key

7.8.2 Exponential Distribution: GATE DA 2025 | Question: 11



It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X . What is the value of λ ? (ln denotes natural logarithm)

- A. $\ln 2$ B. $\ln 4$ C. $\ln 3$ D. $\ln 0.25$

gateda-2025 probability exponential-distribution expectation one-mark

Answer key

7.8.3 Exponential Distribution: GATE DA 2025 | Question: 31



For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as follows

$$\lfloor x \rfloor = k, \quad k \leq x < k + 1,$$

where k is an integer. Let $Y = \lfloor X \rfloor$, where X is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$, where $\ln$ denotes natural logarithm. For any positive integer ℓ , one can write the probability of the event $Y = \ell$ as follows

$$P(Y = \ell) = q^\ell (1 - q)$$

The value of q is

- A. 0.1 B. 0.01 C. 0.5 D. 0.434

gateda-2025 probability random-variable exponential-distribution two-marks

Answer key

7.8.4 Exponential Distribution: GATE DS&AI 2024 | Question: 47



Let X be a random variable exponentially distributed with parameter $\lambda > 0$. The probability density function of X is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

If $5E(X) = \text{Var}(X)$, where $E(X)$ and $\text{Var}(X)$ indicate the expectation and variance of X , respectively, the value of λ is _____ (rounded off to one decimal place).

Answer key

7.8.5 Exponential Distribution: GATE IT 2004 | Question: 33



Let X and Y be two exponentially distributed and independent random variables with mean α and β , respectively. If $Z = \min(X, Y)$, then the mean of Z is given by

- A. $\left(\frac{1}{\alpha + \beta}\right)$ B. $\min(\alpha, \beta)$
 C. $\left(\frac{\alpha\beta}{\alpha + \beta}\right)$ D. $\alpha + \beta$

gateit-2004 probability exponential-distribution random-variable normal

Answer key

7.9

Independent Events (6)

7.9.1 Independent Events: GATE CSE 1994 | Question: 2.8



Let A, B , and C be independent events which occur with probabilities 0.8, 0.5, and 0.3 respectively. The probability of occurrence of at least one of the event is _____

gate1994 probability normal numerical-answers independent-events

Answer key

7.9.2 Independent Events: GATE CSE 1999 | Question: 2.1



Consider two events E_1 and E_2 such that probability of E_1 , $P_r[E_1] = \frac{1}{2}$, probability of E_2 , $P_r[E_2] = \frac{1}{3}$, and probability of E_1 and E_2 , $P_r[E_1 \text{ and } E_2] = \frac{1}{5}$. Which of the following statements is/are true?

- A. $P_r[E_1 \text{ or } E_2]$ is $\frac{2}{3}$ B. Events E_1 and E_2 are independent
 C. Events E_1 and E_2 are not independent D. $P_r[E_1 | E_2] = \frac{4}{5}$

gate1999 probability normal independent-events

Answer key

7.9.3 Independent Events: GATE CSE 2000 | Question: 2.2



E_1 and E_2 are events in a probability space satisfying the following constraints:

- $Pr(E_1) = Pr(E_2)$
- $Pr(E_1 \cup E_2) = 1$
- E_1 and E_2 are independent

The value of $Pr(E_1)$, the probability of the event E_1 , is

- A. 0 B. $\frac{1}{4}$ C. $\frac{1}{2}$ D. 1

gatecse-2000 probability easy independent-events

Answer key

7.9.4 Independent Events: GATE CSE 2023 | Question: 43



Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?

- A. A and B are independent. B. A and C are independent.
 C. B and C are independent. D. $\text{Prob}(B | C) = \text{Prob}(B)$

gatecse-2023 probability independent-events multiple-selects two-marks

Answer key

7.9.5 Independent Events: GATE DA 2025 | Question: 44



Consider a coin-toss experiment where the probability of head showing up is p . In the i^{th} coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

where n is the total number of independent coin tosses.

Which of the following statements is/are correct?

A. $E[\hat{p}] = p$

B. $E[\hat{p}] = \frac{p}{n}$

C. As n increases, variance of $\hat{p}$ decreases

D. Variance of $\hat{p}$ does not depend on n

gateda-2025 probability independent-events multiple-selects two-marks

Answer key

7.9.6 Independent Events: GATE DS&AI 2024 | Question: 2



Three fair coins are tossed independently. T is the event that two or more tosses result in heads. S is the event that two or more tosses result in tails.

What is the probability of the event $T \cap S$?

A. 0

B. 0.5

C. 0.25

D. 1

gate-ds-ai-2024 probability independent-events one-mark

Answer key

7.10

Normal Distribution (2)

7.10.1 Normal Distribution: GATE CSE 2008 | Question: 29



Let X be a random variable following normal distribution with mean $+1$ and variance 4 . Let Y be another normal variable with mean -1 and variance unknown. If $P(X \leq -1) = P(Y \geq 2)$, the standard deviation of Y is

A. 3

B. 2

C. $\sqrt{2}$

D. 1

gatecse-2008 random-variable normal-distribution probability normal

Answer key

7.10.2 Normal Distribution: GATE CSE 2017 Set 1 | Question: 19



Let X be a Gaussian random variable with mean 0 and variance σ^2 . Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b . The median of Y is _____.

gatecse-2017-set1 probability numerical-answers normal-distribution

Answer key

7.11

Poisson Distribution (4)

7.11.1 Poisson Distribution: GATE CSE 1989 | Question: 4-viii



$P_n(t)$ is the probability of n events occurring during a time interval t . How will you express $P_0(t+h)$ in terms of $P_0(h)$, if $P_0(t)$ has stationary independent increments? (Note: $P_t(t)$ is the probability density function).

gate1989 descriptive probability poisson-distribution

Answer key

7.11.2 Poisson Distribution: GATE CSE 2013 | Question: 2



Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3 . What is the probability of observing fewer than 3 cars during any given minute in this interval?

A. $\frac{8}{(2e^3)}$

B. $\frac{9}{(2e^3)}$

C. $\frac{17}{(2e^3)}$

D. $\frac{26}{(2e^3)}$

gatecse-2013 probability poisson-distribution normal

Answer key

7.11.3 Poisson Distribution: GATE CSE 2017 Set 2 | Question: 48



If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(x+2)^2]$ equals ____.

gatecse-2017-set2 expectation poisson-distribution numerical-answers probability

Answer key

7.11.4 Poisson Distribution: GATE IT 2007 | Question: 57



In a multi-user operating system on an average, 20 requests are made to use a particular resource per hour. The arrival of requests follows a Poisson distribution. The probability that either one, three or five requests are made in 45 minutes is given by :

A. $6.9 \times 10^6 \times e^{-20}$

B. $1.02 \times 10^6 \times e^{-20}$

C. $6.9 \times 10^3 \times e^{-20}$

D. $1.02 \times 10^3 \times e^{-20}$

gateit-2007 probability poisson-distribution normal

Answer key

7.12

Probability (28)

7.12.1 Probability: GATE CSE 1995 | Question: 1.18



The probability that a number selected at random between 100 and 999 (both inclusive) will not contain the digit 7 is:

A. $\frac{16}{25}$

B. $\left(\frac{9}{10}\right)^3$

C. $\frac{27}{75}$

D. $\frac{18}{25}$

gate1995 probability normal

Answer key

7.12.2 Probability: GATE CSE 1995 | Question: 2.14



A bag contains 10 white balls and 15 black balls. Two balls are drawn in succession. The probability that one of them is black and the other is white is:

A. $\frac{2}{3}$

B. $\frac{4}{5}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

gate1995 probability normal

Answer key

7.12.3 Probability: GATE CSE 1996 | Question: 1.5



Two dice are thrown simultaneously. The probability that at least one of them will have 6 facing up is

A. $\frac{1}{36}$

B. $\frac{1}{3}$

C. $\frac{25}{36}$

D. $\frac{11}{36}$

gate1996 probability easy

Answer key

7.12.4 Probability: GATE CSE 1996 | Question: 2.7



The probability that top and bottom cards of a randomly shuffled deck are both aces is

A. $\frac{4}{52} \times \frac{4}{52}$

B. $\frac{4}{52} \times \frac{3}{52}$

C. $\frac{4}{52} \times \frac{3}{51}$

D. $\frac{4}{52} \times \frac{4}{51}$

gate1996 probability easy

Answer key

7.12.5 Probability: GATE CSE 1997 | Question: 1.1

The probability that it will rain today is 0.5. The probability that it will rain tomorrow is 0.6. The probability that it will rain either today or tomorrow is 0.7. What is the probability that it will rain today and tomorrow?

- A. 0.3 B. 0.25 C. 0.35 D. 0.4

gate1997 probability easy

Answer key

7.12.6 Probability: GATE CSE 1998 | Question: 1.1

A die is rolled three times. The probability that exactly one odd number turns up among the three outcomes is

- A. $\frac{1}{6}$ B. $\frac{3}{8}$ C. $\frac{1}{8}$ D. $\frac{1}{2}$

gate1998 probability easy

Answer key

7.12.7 Probability: GATE CSE 2001 | Question: 2.4

Seven (distinct) car accidents occurred in a week. What is the probability that they all occurred on the same day?

- A. $\frac{1}{7^7}$ B. $\frac{1}{7^6}$ C. $\frac{1}{2^7}$ D. $\frac{7}{2^7}$

gatecse-2001 probability normal

Answer key

7.12.8 Probability: GATE CSE 2004 | Question: 25

If a fair coin is tossed four times. What is the probability that two heads and two tails will result?

- A. $\frac{3}{8}$ B. $\frac{1}{2}$ C. $\frac{5}{8}$ D. $\frac{3}{4}$

gatecse-2004 probability easy

Answer key

7.12.9 Probability: GATE CSE 2010 | Question: 26

Consider a company that assembles computers. The probability of a faulty assembly of any computer is p . The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q . What is the probability of a computer being declared faulty?

- A. $pq + (1-p)(1-q)$ B. $(1-q)p$ C. $(1-p)q$ D. pq

gatecse-2010 probability easy

Answer key

7.12.10 Probability: GATE CSE 2010 | Question: 27

What is the probability that divisor of 10^{99} is a multiple of 10^{96} ?

- A. $\left(\frac{1}{625}\right)$ B. $\left(\frac{4}{625}\right)$ C. $\left(\frac{12}{625}\right)$ D. $\left(\frac{16}{625}\right)$

gatecse-2010 probability normal

Answer key

7.12.11 Probability: GATE CSE 2011 | Question: 34

A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

A. $\left(\frac{1}{5}\right)$

B. $\left(\frac{4}{25}\right)$

C. $\left(\frac{1}{4}\right)$

D. $\left(\frac{2}{5}\right)$

gatecse-2011 probability normal

Answer key

7.12.12 Probability: GATE CSE 2014 Set 1 | Question: 48



Four fair six-sided dice are rolled. The probability that the sum of the results being 22 is $\frac{X}{1296}$. The value of X is _____.

gatecse-2014-set1 probability numerical-answers normal

Answer key

7.12.13 Probability: GATE CSE 2014 Set 2 | Question: 1



The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p . Then $100p =$ _____.

gatecse-2014-set2 probability numerical-answers normal

Answer key

7.12.14 Probability: GATE CSE 2014 Set 2 | Question: 48



The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is _____.

gatecse-2014-set2 probability numerical-answers normal

Answer key

7.12.15 Probability: GATE CSE 2014 Set 3 | Question: 48



Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the event, the maximum value of $P(A)P(B)$ is _____.

gatecse-2014-set3 probability numerical-answers normal

Answer key

7.12.16 Probability: GATE CSE 2016 Set 1 | Question: 29



Consider the following experiment.

Step 1. Flip a fair coin twice.

Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places)

gatecse-2016-set1 probability normal numerical-answers

Answer key

7.12.17 Probability: GATE CSE 2018 | Question: 15



Two people, P and Q , decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q . Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____.

gatecse-2018 probability normal numerical-answers one-mark

Answer key

7.12.18 Probability: GATE CSE 2021 Set 2 | Question: 33



A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?

- A. $\frac{r}{r+b}$
- B. $\frac{r}{r+b+3}$
- C. $\frac{r+3}{r+b+3}$
- D. $\left(\frac{r}{r+b}\right)\left(\frac{r+1}{r+b+1}\right)\left(\frac{r+2}{r+b+2}\right)\left(\frac{r+3}{r+b+3}\right)$

gatecse-2021-set2 probability normal two-marks

Answer key

7.12.19 Probability: GATE CSE 2024 | Set 1 | Question: 17



Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statements is/are TRUE?

- A. The two events A and B are independent
- B. $P(A \cup B) = 0.7$
- C. $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B
- D. $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B , respectively

gatecse2024-set1 multiple-selects probability one-mark

Answer key

7.12.20 Probability: GATE CSE 2024 | Set 2 | Question: 8



When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (*i.e.*, 1, 2, 3, 4, 5, and 6) is

- A. $\frac{1}{324}$
- B. $\frac{5}{324}$
- C. $\frac{7}{324}$
- D. $\frac{11}{324}$

gatecse2024-set2 probability one-mark

Answer key

7.12.21 Probability: GATE CSE 2025 | Set 1 | Question: 22



A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(\text{head}) = 0.5$ and for a fake coin, $P(\text{head}) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (rounded off to two decimal places)

gatecse2025-set1 probability numerical-answers one-mark

Answer key

7.12.22 Probability: GATE DA 2025 | Question: 35



A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event A represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$

- A. 0
- B. 1
- C. $1 - \left(\frac{5}{6}\right)^{100}$
- D. $\left(\frac{5}{6}\right)^{100}$

gateda-2025 probability two-marks

Answer key

7.12.23 Probability: GATE DS&AI 2024 | Question: 48



Consider two events T and S . Let $\bar{T}$ denote the complement of the event T . The probability associated with different events are given as follows:

$$P(\bar{T}) = 0.6, \quad P(S | T) = 0.3, \quad P(S | \bar{T}) = 0.6$$

Then, $P(T | S)$ is _____ (rounded off to two decimal places).

gate-ds-ai-2024 numerical-answers probability two-marks

Answer key

7.12.24 Probability: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 7



A fair coin is flipped twice and it is known that at least one tail is observed. The probability of getting two tails is

- A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. $\frac{1}{4}$

gateda-sample-paper-2024 engineering-mathematics probability

Answer key

7.12.25 Probability: GATE IT 2004 | Question: 1



In a population of N families, 50% of the families have three children, 30% of the families have two children and the remaining families have one child. What is the probability that a randomly picked child belongs to a family with two children?

- A. $\left(\frac{3}{23}\right)$ B. $\left(\frac{6}{23}\right)$ C. $\left(\frac{3}{10}\right)$ D. $\left(\frac{3}{5}\right)$

gateit-2004 probability normal

Answer key

7.12.26 Probability: GATE IT 2005 | Question: 1



A bag contains 10 blue marbles, 20 green marbles and 30 red marbles. A marble is drawn from the bag, its colour recorded and it is put back in the bag. This process is repeated 3 times. The probability that no two of the marbles drawn have the same colour is

- A. $\left(\frac{1}{36}\right)$ B. $\left(\frac{1}{6}\right)$ C. $\left(\frac{1}{4}\right)$ D. $\left(\frac{1}{3}\right)$

gateit-2005 probability normal

Answer key

7.12.27 Probability: GATE IT 2008 | Question: 2



A sample space has two events A and B such that probabilities $P(A \cap B) = \frac{1}{2}$, $P(A') = \frac{1}{3}$, $P(B') = \frac{1}{3}$. What is $P(A \cup B)$?

- A. $\left(\frac{11}{12}\right)$ B. $\left(\frac{10}{12}\right)$ C. $\left(\frac{9}{12}\right)$ D. $\left(\frac{8}{12}\right)$

gateit-2008 probability easy

Answer key

7.12.28 Probability: GATE IT 2008 | Question: 23



What is the probability that in a randomly chosen group of r people at least three people have the same birthday?

- A. $1 - \frac{365 - 364 \dots (365 - r + 1)}{365^r}$

- B. $\frac{365 \cdot 364 \dots (365 - r + 1)}{365^r} + {}^r C_1 \cdot 365 \cdot \frac{364 \cdot 363 \dots (364 - (r - 2) + 1)}{364^{r+2}}$
- C. $1 - \frac{365 \cdot 364 \dots (365 - r + 1)}{365^r} - {}^r C_2 \cdot 365 \cdot \frac{364 \cdot 363 \dots (364 - (r - 2) + 1)}{364^{r-2}}$
- D. $\frac{365 \cdot 364 \dots (365 - r + 1)}{365^r}$

gateit-2008 probability normal

Answer key

7.13 Probability Density Function (1)

7.13.1 Probability Density Function: GATE CSE 2003 | Question: 60, ISRO2007-45



A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

- A. $f_1(t) + f_2(t)$
- B. $\int_0^t f_1(x)f_2(x)dx$
- C. $\int_0^t f_1(x)f_2(t-x)dx$
- D. $\max\{f_1(t), f_2(t)\}$

gatecse-2003 probability normal isro2007 probability-density-function

Answer key

7.14 Probability Distribution (1)

7.14.1 Probability Distribution: GATE CSE 2025 | Set 1 | Question: 48



Consider a probability distribution given by the density function $P(x)$.

$$P(x) = \begin{cases} Cx^2, & \text{for } 1 \leq x \leq 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$$

The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is _____. (rounded off to three decimal places)

gatecse2025-set1 probability probability-distribution numerical-answers two-marks

Answer key

7.15 Random Variable (8)

7.15.1 Random Variable: GATE CSE 2005 | Question: 12, ISRO2009-64



Let $f(x)$ be the continuous probability density function of a random variable x , the probability that $a < x \leq b$, is :

- A. $f(b-a)$
- B. $f(b) - f(a)$
- C. $\int_a^b f(x)dx$
- D. $\int_a^b xf(x)dx$

gatecse-2005 probability random-variable easy isro2009

Answer key

7.15.2 Random Variable: GATE CSE 2011 | Question: 33



Consider a finite sequence of random values $X = [x_1, x_2, \dots, x_n]$. Let μ_x be the mean and σ_x be the standard deviation of X . Let another finite sequence Y of equal length be derived from this as $y_i = a * x_i + b$, where a and b are positive constants. Let μ_y be the mean and σ_y be the standard deviation of this sequence.

Which one of the following statements is **INCORRECT**?

- A. Index position of mode of X in X is the same as the index position of mode of Y in Y
- B. Index position of median of X in X is the same as the index position of median of Y in Y
- C. $\mu_y = a\mu_x + b$
- D. $\sigma_y = a\sigma_x + b$

Answer key

7.15.3 Random Variable: GATE CSE 2012 | Question: 21



Consider a random variable X that takes values $+1$ and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and $+1$ are

- A. 0 and 0.5 B. 0 and 1 C. 0.5 and 1 D. 0.25 and 0.75

Answer key

7.15.4 Random Variable: GATE CSE 2015 Set 3 | Question: 37



Suppose X_i for $i = 1, 2, 3$ are independent and identically distributed random variables whose probability mass functions are $Pr[X_i = 0] = Pr[X_i = 1] = \frac{1}{2}$ for $i = 1, 2, 3$. Define another random variable $Y = X_1 X_2 \oplus X_3$, where $\oplus$ denotes XOR. Then $Pr[Y = 0 | X_3 = 0] = \underline{\hspace{2cm}}$.

Answer key

7.15.5 Random Variable: GATE DA 2025 | Question: 29



Consider the cumulative distribution function (CDF) of a random variable X :

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases}$$

The value of $P(X^2 \leq 0.25)$ is

- A. 0.625 B. 0.25 C. 0.5 D. 0.5625

Answer key

7.15.6 Random Variable: GATE DA 2025 | Question: 9



Let X be a continuous random variable whose cumulative distribution function (CDF) $F_X(x)$, for some t , is given as follows:

$$F_X(x) = \begin{cases} 0 & x \leq t \\ \frac{x-t}{4-t} & t \leq x \leq 4 \\ 1 & x \geq 4 \end{cases}$$

If the median of X is 3, then what is the value of t ?

- A. 2 B. 1 C. -1 D. 0

Answer key

7.15.7 Random Variable: GATE DS&AI 2024 | Question: 1



Consider the following statements:

- The mean and variance of a Poisson random variable are equal.
- For a standard normal random variable, the mean is zero and the variance is one.

Which **ONE** of the following options is **correct**?

- A. Both (i) and (ii) are true
C. (ii) is true and (i) is false
B. (i) is true and (ii) is false
D. Both (i) and (ii) are false

gate-ds-ai-2024 probability random-variable one-mark

Answer key

7.15.8 Random Variable: GATE DS&AI 2024 | Question: 55



Two fair coins are tossed independently. X is a random variable that takes a value of 1 if both tosses are heads and 0 otherwise. Y is a random variable that takes a value of 1 if at least one of the tosses is heads and 0 otherwise.

The value of the covariance of X and Y is _____ (rounded off to three decimal places).

gate-ds-ai-2024 probability random-variable numerical-answers two-marks

Answer key

7.16 Square Invariant (1)

7.16.1 Square Invariant: GATE CSE 2025 | Set 2 | Question: 54



A quadratic polynomial $(x - \alpha)(x - \beta)$ over complex numbers is said to be square invariant if $(x - \alpha)(x - \beta) = (x - \alpha^2)(x - \beta^2)$. Suppose from the set of all square invariant quadratic polynomials we choose one at random.

The probability that the roots of the chosen polynomial are equal is _____. (rounded off to one decimal place)

gatecse2025-set2 probability square-invariant numerical-answers two-marks

Answer key

7.17 Statistics (2)

7.17.1 Statistics: GATE CSE 2021 Set 2 | Question: 22



For a given biased coin, the probability that the outcome of a toss is a head is 0.4. This coin is tossed 1,000 times. Let X denote the random variable whose value is the number of times that head appeared in these 1,000 tosses. The standard deviation of X (rounded to 2 decimal place) is _____

gatecse-2021-set2 numerical-answers probability random-variable one-mark statistics

Answer key

7.17.2 Statistics: GATE CSE 2024 | Set 2 | Question: 34



Let x and y be random variables, not necessarily independent, that take real values in the interval $[0, 1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$, respectively. Which one of the following statements is TRUE?

- A. $\bar{z} = \bar{x}\bar{y}$
C. $\bar{z} \geq \bar{x}\bar{y}$
B. $\bar{z} \leq \bar{x}\bar{y}$
D. $\bar{z} \leq \bar{x}$

gatecse2024-set2 probability random-variable statistics two-marks

Answer key

7.18 Uniform Distribution (10)

7.18.1 Uniform Distribution: GATE CSE 1998 | Question: 3a



Two friends agree to meet at a park with the following conditions. Each will reach the park between 4:00 pm and 5:00 pm and will see if the other has already arrived. If not, they will wait for 10 minutes or the end of the hour whichever is earlier and leave. What is the probability that the two will not meet?

gate1998 probability normal numerical-answers uniform-distribution

Answer key

7.18.2 Uniform Distribution: GATE CSE 2004 | Question: 78



Two n bit binary strings, S_1 and S_2 are chosen randomly with uniform probability. The probability that the Hamming distance between these strings (the number of bit positions where the two strings differ) is equal to d is

- A. $\frac{nC_d}{2^n}$ B. $\frac{nC_d}{2^d}$ C. $\frac{d}{2^n}$ D. $\frac{1}{2^d}$

gatecse-2004 probability normal uniform-distribution

Answer key

7.18.3 Uniform Distribution: GATE CSE 2004 | Question: 80



A point is randomly selected with uniform probability in the $X - Y$ plane within the rectangle with corners at $(0,0)$, $(1,0)$, $(1,2)$ and $(0,2)$. If p is the length of the position vector of the point, the expected value of p^2 is

- A. $\left(\frac{2}{3}\right)$ B. 1 C. $\left(\frac{4}{3}\right)$ D. $\left(\frac{5}{3}\right)$

gatecse-2004 probability uniform-distribution expectation normal

Answer key

7.18.4 Uniform Distribution: GATE CSE 2007 | Question: 24



Suppose we uniformly and randomly select a permutation from the $20!$ permutations of $1, 2, 3, \dots, 20$. What is the probability that 2 appears at an earlier position than any other even number in the selected permutation?

- A. $\left(\frac{1}{2}\right)$ B. $\left(\frac{1}{10}\right)$ C. $\left(\frac{9!}{20!}\right)$ D. None of these

gatecse-2007 probability easy uniform-distribution

Answer key

7.18.5 Uniform Distribution: GATE CSE 2014 Set 1 | Question: 2



Suppose you break a stick of unit length at a point chosen uniformly at random. Then the expected length of the shorter stick is _____.

gatecse-2014-set1 probability uniform-distribution expectation numerical-answers normal

Answer key

7.18.6 Uniform Distribution: GATE CSE 2019 | Question: 47



Suppose Y is distributed uniformly in the open interval $(1,6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) _____

gatecse-2019 numerical-answers engineering-mathematics probability uniform-distribution two-marks

Answer key

7.18.7 Uniform Distribution: GATE CSE 2020 | Question: 45



For $n > 2$, let $a \in \{0,1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0,1\}^n$. Then, the probability that $\sum_{i=1}^n a_i x_i$ is an odd number is _____

gatecse-2020 numerical-answers probability uniform-distribution two-marks

Answer key

7.18.8 Uniform Distribution: GATE CSE 2024 | Set 1 | Question: 4



Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \dots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

- A. The events X and Y are mutually exclusive B. The events X and Y are independent
C. Either event X or Y must occur D. Event X is more likely than event Y

gatecse2024-set1 probability uniform-distribution one-mark

Answer key

7.18.9 Uniform Distribution: GATE CSE 2025 | Set 2 | Question: 55



The unit interval $(0,1)$ is divided at a point chosen uniformly distributed over $(0,1)$ in $\mathbb{R}$ into two disjoint subintervals.

The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)

gatecse2025-set2 probability uniform-distribution numerical-answers two-marks

Answer key

7.18.10 Uniform Distribution: GATE DS&AI 2024 | Question: 46



Let X be a random variable uniformly distributed in the interval $[1,3]$ and Y be a random variable uniformly distributed in the interval $[2,4]$. If X and Y are independent of each other, the probability $P(X \geq Y)$ is _____ (rounded off to three decimal places).

gate-ds-ai-2024 numerical-answers random-variable probability uniform-distribution two-marks

Answer key

7.19 Variance (1)

7.19.1 Variance: GATE DA 2025 | Question: 26



Let $Y = Z^2, Z = \frac{X - \mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2 . The variance of Y is

- A. 1 B. 2 C. 3 D. 4

gateda-2025 probability random-variable variance two-marks

Answer Keys

7.1.1	0.25:0.25	7.2.1	C	7.2.2	0.125	7.3.1	A	7.4.1	C
7.4.2	D	7.4.3	A	7.4.4	A	7.4.5	A	7.4.6	C
7.5.1	D	7.5.2	N/A	7.5.3	D	7.5.4	A	7.5.5	C
7.5.6	B	7.5.7	A	7.5.8	B	7.5.9	0.55	7.5.10	A
7.5.11	0.60 : 0.61	7.5.12	0.04 : 0.04	7.5.13	0.375	7.5.14	C	7.6.1	0.5
7.7.1	C	7.7.2	D	7.7.3	D	7.7.4	C	7.7.5	C
7.7.6	3.8 : 3.9	7.7.7	B	7.7.8	D	7.7.9	D	7.7.10	C
7.7.11	A	7.7.12	66.6:66.7	7.7.13	C	7.8.1	0.35 : 0.39	7.8.2	A
7.8.3	A	7.8.4	0.2	7.8.5	C	7.9.1	0.93	7.9.2	C
7.9.3	D	7.9.4	C;D	7.9.5	A;C	7.9.6	A	7.10.1	A
7.10.2	0	7.11.1	N/A	7.11.2	C	7.11.3	54	7.11.4	B
7.12.1	D	7.12.2	C	7.12.3	D	7.12.4	C	7.12.5	D
7.12.6	B	7.12.7	B	7.12.8	A	7.12.9	A	7.12.10	A
7.12.11	A	7.12.12	10	7.12.13	11.85 : 11.95	7.12.14	0.259 : 0.261	7.12.15	0.25
7.12.16	0.33 : 0.34	7.12.17	0.0230 : 0.0232	7.12.18	A	7.12.19	B;C	7.12.20	B
7.12.21	0.49:0.51	7.12.22	C	7.12.23	0.25	7.12.24	B	7.12.25	B
7.12.26	B	7.12.27	B	7.12.28	X	7.13.1	C	7.14.1	0.300:0.302
7.15.1	C	7.15.2	D	7.15.3	C	7.15.4	0.75	7.15.5	C

7.15.6	A
7.17.2	D
7.18.5	0.24 : 0.27
7.18.10	0.125

7.15.7	A
7.18.1	0.69:0.70
7.18.6	0.8
7.19.1	B

7.15.8	0.062:0.063
7.18.2	A
7.18.7	0.5

7.16.1	0.5:0.5
7.18.3	D
7.18.8	B

7.17.1	15.00 : 16.00
7.18.4	B
7.18.9	0.70:0.80